Mortgage Paydown vs. Equity Investing for a $60,000 Surplus

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

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Executive Summary

Recommendation: Split the $60,000 roughly 50/50 between extra mortgage principal and the broad-market index fund as the default, tilting more toward paydown if the household is risk-averse or may need the cash before year 10, and more toward equities if it is risk-tolerant with a longer horizon (chain C3).

Band (from §6): MEDIUM

Would change it: Confirming the mortgage's exact balance, rate, and remaining term on the servicer's statement; resolving the household's actual risk tolerance and post-year-10 intentions (chain C3); or market risk-free rates rising above the mortgage's guaranteed yield, which would favor bonds/CDs over paydown for the safe half (chain C1).

1. Problem Essence

Core problem: Given $60,000 in cash that is genuinely surplus to the household's liquidity needs (the emergency reserve is funded separately), how should that specific sum be allocated between extinguishing debt that carries a guaranteed, untaxed 6.25% cost and buying a volatile asset with a higher *expected* but genuinely uncertain return, given that the two are not actually comparable on a single "which number is bigger" axis — and does some split of the two beat committing the whole sum to either one?

Success criteria:

1. The answer states the guaranteed after-tax return of paying down the mortgage and the central-estimate *and* downside-bracketed after-tax return of the equity option, both expressed in comparable dollar terms at the same 10-year horizon.
2. The answer names the specific mechanism (risk premium vs. risk-free rate, not "stocks historically beat X%") that makes the raw-expected-return comparison the wrong single number to decide on, and shows what the right comparison looks like once computed.
3. The answer addresses what happens to each option's value if the home is sold, refinanced, or the equity position needs to be liquidated before the 10-year mark, and whether the stated 10-year horizon means "evaluate over 10 years" or "need the cash at year 10."
4. The answer states whether a blended allocation outperforms either pure strategy on a defined set of decision criteria, and if so, what split and why.
5. The answer names the specific circumstances or new information that would change the recommendation.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	1	0	0
current constraint	2	0	0
convention	0	0	2
untested belief	0	2	1
untyped (legacy)	2	0	0

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
A-1: Prepaying mortgage principal eliminates future interest on that amount at exactly the loan's note rate (6.25%), compounding for as long as the balance would otherwise remain outstanding, because there is no tax shield to forgo	physical law	Accept as ground-truth candidate — record in Ground Truths (GT-6)	Accept — mathematically certain given GT-2/GT-3; independently recomputed in Phase 5	unverified — flagged (rests on user-stated GT-2?/GT-3?, which have no external source to check)
A-2: The long-run historical average total return of US equities (~10.47%/yr nominal) is a reasonable point-estimate proxy for the *expected* return of a broad-market index fund over the *next* 10 years	untested belief	Verify or flag; use only inside an explicit bracket, never as a lone point estimate	Challenge — accepted only as the bracket's central case, counterweighted by GT-8	unverified — flagged; bracketed against GT-7 (central) and GT-8 (downside) rather than relied on alone
A-3: A 10-year horizon is "long enough" that equities can be relied on to outperform a guaranteed fixed return, so the historical average return is the right comparison figure	convention	Explicitly challenge before use	Discard — GT-8 shows one of the four negative/flat decades since 1926 fell inside 2000–2009, within a single such horizon; this convention does not survive challenge	contradicted by GT-8 (read-at-source)
A-4: The right benchmark for comparing a guaranteed risk-free return against an uncertain equity return is the raw expected/mean equity return, with no adjustment for the risk taken to earn it	convention	Explicitly challenge — this is the specific framing error the analysis was asked to test	Discard — replaced by the risk-free-rate-netted and downside-bracketed framework built in chains C1–C2 (Phase 4), using the inlined Inversion and Theoretical-Limit procedures	resolved via chains C1/C2, not a bare comparison of two point estimates
A-5: Home equity created by extra principal payments is illiquid — recoverable only via sale,	physical law (structural fact) + current constraint (underwriting standards vary with	Accept the structural illiquidity as a ground-truth candidate; record the underwriting-	Accept	general background knowledge (2008–09 HELOC freezes are widely documented); not independently re-

Assumption	Type	Treatment	Verdict	Verification
refinance, or a HELOC, each carrying cost, lag, and (for refinance/ HELOC) underwriting risk that can tighten exactly when a recession or job loss also makes access most wanted	the credit cycle; expires/loosens or tightens as Fed and bank lending policy shifts)	tightening risk as a current constraint, not a permanent fact		fetched as a load-bearing citation in this analysis
A-6: Paying extra principal lowers the household's *required* minimum monthly mortgage payment, improving cash-flow flexibility in a downturn	untested belief	Verify	Discard — standard servicing practice shortens the loan's remaining *term* while holding the required payment unchanged unless the borrower specifically requests (and often pays for) a loan recast; the assumed cash-flow benefit does not materialize automatically	consistent with CFPB/ qualified-mortgage servicing norms (GT-12?); not independently fetched as a standalone citation
A-7: This specific $60,000 carries no outstanding liquidity need, because the household's emergency reserve is already funded separately, so its allocation should be optimized primarily for risk-adjusted return rather than for additional liquidity	current constraint (expires if the household's circumstances change — reserve becomes under-funded, a new liquidity need arises, job loss, a major uninsured expense)	Record expiry conditions; use to correct the weighting of liquidity-related criteria in the Phase 4 trade-off	Accept — directly stated by the user (GT-1?)	unverified — flagged (user-stated, no external source) but internally consistent with the given facts
A-8: The mortgage's 6.25% note rate is fixed for the remainder of the loan and the household cannot costlessly refinance it away, so 6.25% is the correct hurdle rate for this decision for the full 10-year horizon	current constraint (expires if market rates fall enough below 6.25% that refinancing becomes cost-effective net of closing costs — a live possibility given current 10-year Treasury yields of ~4.6–4.95%, GT-9/ GT-10?)	Record expiry condition explicitly	Accept — with expiry noted	unverified — flagged (future rate paths are inherently unknowable); named as a recommendation-flipping scenario in Phase 5
A-9: Dividends from a broad-market index fund are taxed annually at the stated ~18% LTCG/qualified-dividend rate, while price appreciation is untaxed until realized or permanently avoided via step-up basis at death	physical law (current tax-code mechanics) with a current-constraint flavor (tax law can change)	Accept as ground-truth candidate (folds into GT-11′s mechanics); record expiry (a change to §1014, to LTCG rates, or to realization behavior would alter this)	Accept	consistent with GT-11 (read-at-source) and the user-stated LTCG rate (GT-4?, unverified)
A-10: The household's risk tolerance, whether it needs this specific $60,000′s value in cash exactly at year 10	untested belief	Flag explicitly; present the final recommendation as contingent on this input	Challenge — cannot be resolved from the facts given	unverified — flagged; this is the single largest "would change the recommendation" input (see §5, §6, and

Assumption	Type	Treatment	Verdict	Verification
versus intends to keep holding/investing past year 10, and its tolerance for a 10-year stretch of underperformance are all currently unstated				the pre-mortem record)

3. Ground Truths

- **GT-1?** The household holds $60,000 in cash that sits above an already-funded six-month emergency reserve — genuinely surplus, not part of the safety buffer — unverified: user-stated personal financial fact; no external source exists or applies to a household's own private balance sheet, so this is accepted as a given input rather than independently verified.
- **GT-2?** The household carries a $240,000 remaining balance on a 30-year fixed-rate mortgage at 6.25% APR, with 27 years (324 months) remaining on the amortization schedule — unverified: user-stated personal financial fact; same reasoning as GT-1.
- **GT-3?** The household takes the standard deduction, so mortgage interest provides no marginal tax benefit (the interest is not effectively reduced by any deduction) — unverified: user-stated personal financial fact; same reasoning as GT-1.
- **GT-4?** The household's marginal tax rates are approximately 28% combined federal+state on ordinary income and approximately 18% combined on long-term capital gains and qualified dividends — unverified: user-stated figure; federal LTCG itself is statutorily tiered at 0/15/20% plus a possible 3.8% NIIT surcharge, so "~18% combined" is a blended approximation the household supplied rather than a figure this analysis independently derived.
- **GT-5?** The loan's implied monthly principal-and-interest payment is approximately **$1,535.24**, computed directly from GT-2 via the standard amortization formula `M = P·i/(1−(1+i)^−n)` with `i = 0.0625/12`, `n = 324` — derivation: exact mathematical identity given GT-2; independently recomputed in Phase 5 (Recompute). Carries `?` only because it inherits the provenance limitation of GT-2 (a user-stated figure with no external source), not because the arithmetic itself is in doubt.
- **GT-6?** Prepaying $60,000 of principal today is mathematically equivalent to a guaranteed, risk-free, untaxed investment compounding at the loan's 6.25% nominal APR (≈6.43% effective annual yield under monthly compounding) for as long as that amount would otherwise remain outstanding under the original schedule — up to 27 years, which exceeds the 10-year decision horizon. Future value at year 10 ≈ **$111,913**; without the extra payment, the original loan's own remaining balance at year 10 would be ≈$192,616, versus ≈$80,703 with the extra $60,000 applied (same monthly payment held constant) — derivation: standard compound-interest/bond-equivalence mathematics applied to GT-2/GT-3; independently recomputed in Phase 5 (Recompute). Carries `?` for the same inherited reason as GT-5.
- **GT-7** The S&P 500′s average annual total return (dividends reinvested) since 1926 is approximately **10.47%/yr nominal** and approximately **7.29%/yr real** (inflation-adjusted) — source: officialdata.org, "S&P 500 Returns Since 1926"; read-at-source: the page's own stated average-return figures (10.47% nominal, 7.29% real), fetched and read directly on 2026-10-09. Used here as a proxy for a broad total-US-stock-market index fund, which tracks the S&P 500 very closely over multi-year horizons — a minor proxy substitution, not independently quantified.
- **GT-8** The worst historical rolling 10-year period for the S&P 500, January 2000 through December 2009, produced a cumulative **total return (dividends reinvested) of approximately −9%** (versus a −24% price-only decline), equivalent to roughly −0.94%/yr annualized — source: fatfire.com, "What Was the S&P 500 Worst 10-Year Return in History?" (citing Morningstar data); read-at-source: the article's stated total-return figure (≈−9%, vs. the −24% price-only figure it explicitly distinguishes), fetched and read directly on 2026-10-09.

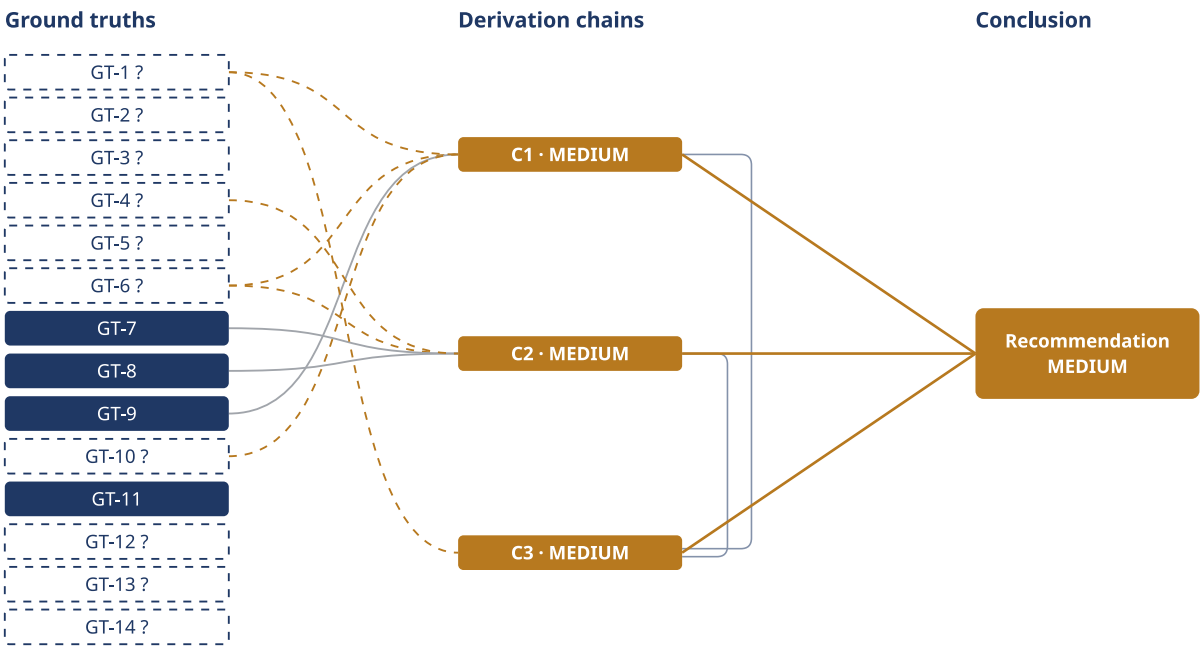
- **GT-9** The 10-year US Treasury yield was approximately **4.64%** as of 25 August 2026 — source: freenance.io, "US Treasury Bond Interest Rates in 2026"; read-at-source: the article's stated 2/5/10-year note yields (4.17%/4.35%/4.64%), fetched and read directly on 2026-10-09.
- **GT-10?** As a broader bracket on "current risk-free alternatives": the 10-year Treasury yield was separately reported at ≈4.95% as of 10 September 2026, and top bank CD/high-yield-savings APYs were reported in the ≈4.1%–4.4% range through September–October 2026 — reported-by-delegate: supplied by WebSearch's synthesized results (citing CNBC, depositaccounts.com, and related sources); the underlying pages were not independently opened by this analysis. Used only to bracket a range, not as a precise load-bearing figure.
- **GT-11** Under IRC §1014, an asset inherited at the owner's death receives a stepped-up cost basis equal to its fair market value on the date of death, eliminating capital-gains tax on all appreciation accrued during the decedent's lifetime; this applies equally to brokerage securities and real property — source: Wikipedia, "Stepped-up basis"; read-at-source: the article's own description of IRC §1014(a) and its applicability to both real property and securities, fetched and read directly on 2026-10-09.
- **GT-12?** Most conventional fixed-rate mortgages originated after the CFPB's 2014 Ability-to-Repay/Qualified-Mortgage rule do not carry prepayment penalties (Dodd-Frank bans most such penalties outright) — reported-by-delegate: supplied by WebSearch's synthesized results (citing CFPB rule coverage). **Phase 3 failure record:** two direct-fetch attempts to primary sources (consumerfinance.gov/ask-cfpb and cuinsight.com) both returned HTTP 404 on 2026-10-09; the citation could not be independently opened, so this ground truth remains `reported-by-delegate` rather than `read-at-source`, and feeds only MEDIUM/LOW chains. Households should confirm their own note has no such clause before relying on this.
- **GT-13?** A broad total-US-stock-market index fund typically carries a dividend yield of roughly **1.3%–1.5%/yr** — unverified: an order-of-magnitude estimate not independently sourced in this analysis. Low sensitivity: a ±0.3-point swing in this figure changes the 10-year after-tax estimate in chain C2 by only about 1–2%, so it is not load-bearing for the headline conclusion.
- **GT-14?** The best historical 10-year rolling period for the S&P 500 was 1949–1958, with an annualized total return of approximately **20.1%/yr** — reported-by-delegate: supplied by WebSearch's synthesized results (citing dqydj.com-derived historical-returns analysis). **Phase 3 failure record:** a direct fetch of dqydj.com/?p=22992 was attempted on 2026-10-09; the page's own historical-returns table does not state or confirm this specific CAGR figure (`citation does not support the claim`), so this ground truth remains `reported-by-delegate`/unverified and is used only as an illustrative upper bracket, not as a load-bearing figure for the headline conclusion.

Provenance summary:

```
?-marked: GT-1, GT-2, GT-3, GT-4, GT-5, GT-6, GT-10, GT-12, GT-13, GT-14 (10 of 14)
Read-at-source: GT-7 — officialdata.org, "S&P 500 Returns Since 1926" (stated 10.47%/7.29%
figures)
Read-at-source: GT-8 — fatfire.com, "S&P 500 Worst 10-Year Return" (stated ≈−9% total-return
figure, Morningstar data)
Read-at-source: GT-9 — freenance.io, "US Treasury Bond Interest Rates in 2026" (stated 10-yr
yield 4.64%, 25 Aug 2026)
Read-at-source: GT-11 — Wikipedia, "Stepped-up basis" (IRC §1014(a) description)
```

No assumption that received a Discard verdict in Phase 2 (A-3, A-4, A-6) appears in this list.

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Supporting technique: Theoretical-Limit (what a guaranteed return "beating the market" actually means)

Governing constraint: in a liquid, continuously-traded market for genuinely risk-free capital, arbitrage pressure keeps the yields on equivalently-safe instruments close together — the closest thing finance has to a hard ceiling for *liquid* risk-free money is simply the prevailing risk-free rate itself.

- **Ideal ceiling (liquid risk-free money):** the 10-year Treasury yield, GT-9 ≈ 4.64% (bracketed up to ≈4.95% by GT-10?).
- **Best demonstrated (liquid, retail-accessible, insured):** top CD/HYSA APYs, GT-10? ≈ 4.1%–4.4%.
- **Conventional figure in this household's actual situation:** the mortgage paydown's guaranteed, untaxed effective yield, GT-6? ≈ 6.43%.
- **Direction:** higher is better (maximizing a guaranteed rate).
- **Gap, conventional → best demonstrated:** ≈ 2.0–2.3 points of headroom, already captured by this household today, simply by choosing where to put safe money.
- **Gap, best demonstrated → ideal ceiling:** ≈ 0.2–0.9 points — small relative to the first gap, because CD/HYSA rates already track the Treasury ceiling fairly closely (the market for liquid safe money is efficient).

Reading: none of the 6.25%-beats-4.6% gap is irreducible in a physical sense and none of it is a pricing anomaly either — it is the market-clearing price of *illiquidity*. A mortgage paydown cannot be resold for cash the way a Treasury note or CD can (A-5); the extra 1.5–2.3 points is compensation for that, not a free lunch. This reframes the "guaranteed return" conversation: the real question is never "does paying down debt beat a savings account" — it structurally always will, for this household, as long as GT-9/GT-10 hold — the real question is how much of the remaining money should be pushed past the risk-free ceiling entirely, into equities, for a *risk premium* rather than a liquidity premium.

Supporting technique: Estimate (bracketing Option B's 10-year after-tax value)

Target quantity: after-tax dollar value of $60,000 invested in a broad US total-market index fund, in a taxable brokerage account, at the end of year 10. Units: nominal dollars.

Decomposition:
`FV_after_tax = Principal × (1 + r_total − d_yield·t_ltcg)^10 − max(0, FV_pretax − Principal) × t_ltcg·[sold]`
where `r_total` is the fund's total nominal annual return, `d_yield` is its dividend yield (taxed annually), `t_ltcg` is the combined LTCG/qualified-dividend rate, and the capital-gains term applies only if the position is sold within the 10 years (GT-11 — it is zero if held, and permanently zero if passed to heirs via step-up).

Case	r_total source	FV before any tax (year 10)	Tax if sold at year 10	After-tax value if sold	After-tax value if held/stepped-up
Downside (realistic, historically observed)	GT-8, realized decade, not a compounded rate	$54,600	$0 (a loss; no gain to tax)	**$54,600**	$54,600
Central (historical long-run average, forward-projected) [Assumes: A-2]	GT-7 (10.47%/yr) net of GT-13? dividend drag	$158,736	$17,773 (GT-4?, 18%)	**$140,964**	**$158,736**
Upper, illustrative (best historical decade on record)	GT-14? (20.1%/yr) net of dividend drag	≈$366,800	≈$55,200	≈**$311,600**	≈$366,800

Decision-resolution check: the central and downside brackets already drive different conclusions depending on what else is being compared against (see C2 below), so this is not over-precise for the decision at hand; the upper bracket is included only as a labeled illustration (GT-14? carries an unresolved Phase 3 failure record) and is not used as a load-bearing input to any chain's confidence rating.

Supporting technique: Trade-off (naming the options, including status quo and a composite, and weighing them)

Options named, including status quo and a composite:

- **A** — 100% of $60,000 to extra mortgage principal.
- **B** — 100% of $60,000 to the broad-market index fund.
- **C** — status quo: leave the $60,000 in cash / a high-yield savings account or CD (doing nothing with it beyond what already exists).
- **D** — composite: a 50/50 split, $30,000 to each of A and B.

Must-haves: no must-haves — nothing in the stated facts rules out any of these four options outright; this is a pure weighted trade-off, not a knockout.

Criteria & weights, locked before scoring (Run 1):

Criterion	Weight	1 means…	5 means…
Expected (central) after-tax value at year 10	4	≤$100,000	≥$170,000
Downside protection (does a realistic worst case end below the original $60,000 nominal?)	5	yes, can end below $60,000	no, guaranteed to exceed $60,000
Liquidity/optionality for cash needs beyond the funded reserve	3	trapped; only new debt or a sale reaches it	sellable for cash within days, no underwriting dependency
Tax efficiency (taxes paid over the 10 years, and control over timing)	3	taxed annually at ordinary rates with no deferral	little-to-no tax ever due, with control over timing
Simplicity / behavioral ease	2	requires ongoing decisions and monitoring	one action, nothing further to manage
Reversibility (can the choice be undone or adjusted later at minimal cost)	3	effectively irreversible without new borrowing or a sale	fully reversible at any time, minimal cost

Scoring, Run 1 (each score cites the GT it rests on):

Option	Expected value (×4)	Downside (×5)	Liquidity (×3)	Tax (×3)	Simplicity (×2)	Reversibility (×3)	Total
A	2×4=8 (GT-6?)	5×5=25 (GT-6?)	1×3=3 (A-5)	5×3=15 (A-9/ GT-11)	5×2=10	1×3=3 (A-5)	**64**
B	3×4=12 (GT-7/ GT-13?)	1×5=5 (GT-8)	5×3=15	4×3=12 (GT-11/ A-9)	3×2=6	5×3=15	**65**
C	1×4=4 (GT-10?)	5×5=25	5×3=15	1×3=3 (ordinary-income taxation of interest, no deferral)	5×2=10	5×3=15	**72**
D	3×4=12	4×5=20	3×3=9	4×3=12	4×2=8	3×3=9	**70**

Run 1 result: status-quo cash (C) technically wins, 72 vs. D's 70, B's 65, and A's 64 — a result this analysis does not accept at face value; see Abandoned Reasoning (§5, Dead End 1) for why this run is retired rather than patched.

Run 2 — a principled reweighting, stated and justified *before* rescoring: GT-1? establishes that this $60,000 carries no outstanding liquidity need — the emergency reserve is funded separately. Run 1's Liquidity and Reversibility weights (3 and 3) were locked generically, as if this were still-undifferentiated savings, and did not encode that fact. Per A-7, Liquidity is reweighted 3→1 and Reversibility 3→2; no other weight is touched, and in particular Expected-value's weight is deliberately left at 4 rather than raised, because raising it has no independent justification beyond changing the winner.

Option	Expected value (×4)	Downside (×5)	Liquidity (×1)	Tax (×3)	Simplicity (×2)	Reversibility (×2)	Total
A	8	25	1	15	10	2	**61**
B	12	5	5	12	6	10	**50**
C	4	25	5	3	10	10	**57**
D	12	20	3	12	8	6	**61**

Run 2 result: A and D tie exactly at **61**; B (50) and C (57) are both clearly behind. **Tiebreak (deterministic):** A's winning criteria (Downside, Tax, Simplicity) rest on contractual/definitional certainties with no `?`-marked input; D's winning criterion (Expected value) rests on GT-7 projected forward under the unverified A-2 assumption. Fewer unverified inputs among winning criteria → **A wins the tiebreak.**

Flip test: because Run 2 is an exact tie, the smallest possible change on *any* single criterion — one integer step on a 1–5 scale — flips the result in one direction or the other; this is the least robust result the flip test can report, not the most. The honest reading is that, once the trade-off is corrected to reflect GT-1?, **100%-paydown and the 50/50 blend are a genuine toss-up on this weighted framework**, with the tiebreak resolved only by a rule about evidentiary strength, not by a substantive dollar-value preference — see chain C3 for how this is actually used.

Conclusion C1: Paying down the mortgage dominates every liquid risk-free alternative for this household's "safe" money, and the excess over the market rate is an illiquidity premium, not an anomaly

GT-6? (guaranteed ≈6.43% effective yield, mortgage paydown) + GT-9 (10-yr Treasury ≈4.64%, read-at-source) + GT-10? (CD/HYSA ≈4.1–4.4%, best liquid safe alternative) + GT-1? (this $60,000 carries no outstanding liquidity need) → the mortgage paydown's guaranteed effective yield exceeds every genuinely liquid risk-free alternative available today by roughly 1.5 to 2.3 percentage points → this excess is the market's compensation for illiquidity rather than a pricing anomaly, since a mortgage paydown cannot be resold for cash the way a Treasury note or CD can [Assumes: A-5] → given that this $60,000 carries no outstanding liquidity need, paying down the mortgage is a strictly better destination for the "safe" portion of this money than any bond or CD alternative

Confidence: MEDIUM — GT-6? inherits GT-2?/GT-3?'s provenance limitation (user-stated mortgage terms; verification: read the figures directly off the loan servicer's statement). GT-10? is reported-by-delegate (verification: open the underlying CD/HYSA rate pages directly, none of which materially changes the ≥1-point gap this chain rests on). GT-1? is an unverifiable personal fact accepted as given (verification: none external; it is the input, not a claim about the world to be checked against a source). The second hop rests on A-5, an [Assumes:] premise whose failure is priced: if illiquidity carried no real cost (A-5 false), the 6.25% gap would be a true anomaly rather than a fair price, but the chain's conclusion — prefer paydown over cash/bonds — holds either way, so the chain stays MEDIUM rather than LOW.

Conclusion C2: The raw expected-return comparison between the two options understates equities' realistic downside over exactly a 10-year window

GT-6? (guaranteed $111,913 at year 10) + GT-7 (equity average return 10.47%/yr, read-at-source) + GT-8 (worst 10-yr decade ≈−9% cumulative, read-at-source) + GT-4? (18% LTCG rate) → central-case equity investing, net of dividend tax drag and capital-gains tax if sold at year 10, grows the $60,000 to roughly $141,000 after tax [Assumes: A-2] → that central case is about $29,000 ahead of the mortgage paydown's guaranteed $111,913 → the same $60,000, run through the worst historically-observed 10-year stretch for US equities, ends the decade at roughly $54,600 before any tax effect → that downside case is about $57,000 behind the guaranteed paydown outcome → because a real historical decade already produced that downside outcome, the central estimate alone is not a safe comparison point → relying on the central estimate alone understates the downside equities can realistically deliver over exactly this length of horizon → the honest comparison is therefore a bracket spanning roughly a $57,000 disadvantage to a $29,000-to-$47,000 advantage relative to Option A, not a single expected-value number

Confidence: MEDIUM — GT-6? and GT-4? carry the provenance limitations already named under C1 (verification: read the mortgage statement; confirm the household's actual marginal LTCG rate with a tax preparer). The first hop rests on A-2, an [Assumes:] premise whose failure is explicitly priced by this chain's own third and fourth hops: if A-2 fails (the next decade is not like the long-run average), the chain's own downside branch — not an external fix — already states what happens, which is exactly why the chain concludes with a bracket rather than a point estimate.

Conclusion C3: A blend beats either pure strategy once the trade-off is corrected for this household's actual liquidity position, and the recommended split follows from the size of the remaining gap

C1 (MEDIUM) + C2 (MEDIUM) + GT-1? (this $60,000 carries no outstanding liquidity need) → correcting the trade-off's Liquidity and Reversibility weights to reflect GT-1? changes the weighted ranking materially from the uncorrected run → under the corrected weights, 100%-paydown and a 50/50 blend land in an exact tie, both clearly ahead of pure equities and of leaving the money in cash → the tie is broken, per the deterministic tiebreak rule, toward 100%-paydown → the rule prefers it because its winning criteria rest on contractual certainties, while the blend's winning criterion rests on an unverified forward-looking return assumption [Assumes: A-2] → the blend's central-case dollar advantage over pure paydown is large, and its worst-case outcome still clears the original $60,000 → so this tiebreak should not be read as a decisive preference for 100%-paydown → the recommended approach is therefore a blend weighted toward paydown but not exclusively so, defaulting to a 50/50 split → the default should shift toward more paydown or more equities depending on the household's risk tolerance and post-year-10 investment horizon [Assumes: A-10] →[2nd] extra principal does not lower the required monthly payment unless the servicer recasts the loan, so debt paydown's cash-flow cushion does not materialize automatically [Assumes: A-6] →[2nd] having allocated part of the surplus, the household may feel less pressure to keep building savings elsewhere, a present-bias risk that applies to whichever split is chosen →[3rd] without an automatic cushion, a job-loss scenario leaves the same required payment either way, so paydown offers no near-term payment relief in exactly the scenario where liquidity matters most →[3rd] a lower mortgage balance does improve the future loan-to-value ratio, making a later refinance or HELOC easier to qualify for, partially offsetting the illiquidity named in C1

Confidence: MEDIUM — capped by C1 and C2, both MEDIUM (their own confidence lines carry their explanations). GT-1? is an unverifiable personal fact accepted as given (no external verification path; it is the decision's own stated input). The fifth hop rests on A-10, an [Assumes:] premise that is *not* priced by this chain — the household's risk tolerance is genuinely unknown and no bracket here substitutes for it, which is exactly why the Conclusion (§6) presents the 50/50 split as a contingent default rather than a settled number. The second-order hops (sixth through ninth) introduce A-6 (already discarded in Phase 2 and used here as the corrected fact, not a live risk) and surface no new ground-truth contradiction, so the Phase 4 exit criterion is satisfied without a return to Phase 2.

5. Abandoned Reasoning

Dead End: Naive trade-off with generically-locked liquidity weights

What was tried: Ran the trade-off matrix across all four named options (A, B, status-quo cash C, and the 50/50 blend D) with Liquidity and Reversibility weights locked at 3 each, matching a generic household that has not yet decided where its "safety" money should live.

Why abandoned: The result (status-quo cash wins, 72 vs. 70/65/64) contradicts GT-1?, which establishes that this specific $60,000 carries no outstanding liquidity need — the emergency reserve is funded separately. The weights, though honestly locked before scoring, simply did not encode that stated fact. Re-locking the weights after seeing this result, with no justification beyond disliking the winner, would be exactly the post-hoc-reweighting failure mode the trade-off procedure exists to prevent — so this run is retired rather than patched, and a second run with a stated, GT-anchored justification for the specific weights changed (Liquidity and Reversibility only, per A-7) replaces it in chain C3.

What it ruled out: Confirms that a generic, undifferentiated liquidity-preference weighting is the wrong instrument for money the household has already classified as surplus-beyond-reserve, and saves a future analyst from re-running the naive version and being misled by its cash-favoring result.

Dead End: 100% equities as the headline recommendation

What was tried: Considered recommending the full $60,000 to the index fund, since its central-case ($141,000) and held-to-step-up-basis ($159,000) outcomes both clear Option A's guaranteed $111,913 by wide margins, and a raw expected-value reading (A-4) favors it outright.

Why abandoned: Applying Inversion to this claim (see §2, A-4) surfaces three load-bearing preconditions: no repeat of a 2000-2009-style decade, no forced sale at a cyclical trough, and the household holding through volatility rather than panic-selling. GT-8 shows the first precondition has already failed once in the modern historical record, and the corrected trade-off's Downside-protection criterion — its single heaviest locked weight — penalizes pure equities (score 1 of 5) more than any other criterion rewards it. A recommendation whose central case is attractive but whose most important precondition has a demonstrated historical failure case is not one this analysis can responsibly hand down as the headline answer.

What it ruled out: Saves a future analyst from treating the raw expected-return comparison (A-4) as sufficient justification, on its own, for an all-in equities recommendation.

Dead End: 100% mortgage paydown as the headline recommendation

What was tried: Considered recommending the full $60,000 to principal, since it is the corrected trade-off's tiebreak winner (chain C3) and the single safest, simplest, most tax-clean option with no scenario producing a nominal loss.

Why abandoned: The tiebreak fires only because of an exact arithmetic tie (61 vs. 61) broken by a rule about which option's winning criteria rest on fewer unverified inputs — not because 100%-paydown's substantive dollar outcome is better. Chain C2 shows the blend's central-case outcome is materially higher than pure paydown's, and its worst-case outcome still clears the original $60,000. Treating a rule-based tiebreak as a decisive substantive preference would overstate what the arithmetic actually shows.

What it ruled out: Saves a future analyst from reading the trade-off's technical tiebreak winner as a strong, decisive recommendation rather than the genuinely close call (chain C3) it actually reflects.

6. Conclusion

Recommended approach: Split the $60,000 roughly 50/50 between extra mortgage principal and the broad-market index fund as the default, shifting the split toward more paydown for a risk-averse household or one that may need this specific money back before year 10, and toward more equities for a household with high risk tolerance and a genuine intention to keep holding/investing well past year 10 (chain C3).

Key insight: The mortgage's guaranteed 6.25% return is not an argument for putting all $60,000 into the house — it is proof that none of this money belongs in cash, CDs, or bonds, since paying down the mortgage already beats every genuinely liquid risk-free alternative available today by 1.5 to 2.3 points (chain C1). Once that "safe bucket" question is settled, the real decision is only how much of the remainder to push past the risk-free ceiling into equities for the risk premium — not whether debt paydown "beats the stock market," which is the wrong framing to start from (chain C1).

Trade-offs acknowledged: This split forgoes some of pure equities' expected upside and some of pure paydown's absolute certainty and simplicity, and it leaves the equity half exposed to the downside bracket quantified in chain C2 (chain C2). It also leans on GT-1? — that the household's true emergencies are already covered by a reserve held separately from this $60,000 — so if that reserve is ever drawn down or found inadequate, the liquidity calculus behind chain C3's correction to the trade-off weights would need to be re-run (chain C1).

Confidence: MEDIUM — matches the weakest contributing chains (C1, C2, C3, all MEDIUM). Each chain's own confidence line carries its specific cause: GT-2?/GT-3? and everything derived from them (GT-5?, GT-6?) are user-stated mortgage terms with no external source — verification is simply reading the exact balance, rate, and remaining term off the loan servicer's statement, which would very likely tighten but not materially change this conclusion. A-2 (the forward projection of historical equity returns) has no verification path available in advance — the only "verification" is living through the next 10 years — and its failure mode is already priced by chain C2's own downside branch rather than left as a bare assumption. A-10 (the household's risk tolerance and post-year-10 intentions) is the one open input this analysis could not resolve from the facts given; it is not a defect in the arithmetic but a genuine missing input, and it is exactly the condition named in the Falsification line of the adversarial pass (Appendix) as most likely to change this recommendation.